# Modeling and Simulation of Mineral Processing Plants

## An Introduction to the Use of Plant Simulation
## Assignment Solutions

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Assignment 3-1. Suggestions for solution.

The flowsheets are easy to simulate using MODSIM. Some results from the simulation of Flowsheet C are shown below and the job is available as a packed job for downloading from the Internet. You should check whether the crushers and screens that have been specified will handle the tonnages that are indicated. Using the simple classical screen model, the data shown in Table 1 were obtained.

Table 1

| Screen | Primary | Secondary | Tertiary |
|---|---|---|---|
| Area required | 17.2 $m^2$ | 7.69 $m^2$ | 4.11 $m^2$ |
| Area installed (top) | | 8.72 $m^2$ | 14.7 $m^2$ |
| Area required | | 18.5 $m^2$ | 4.65 $m^2$ |
| Area installed (bottom) | | 8.72 $m^2$ | 14.7 $m^2$ |

All screens, with the exception of the lower deck or the secondary should handle the loads satisfactorily. The simulation suggests that two screens are not required for the tertiary circuit – one should be sufficient.

The capacities of the crushers are shown in Table 2.

Table 2

| | Brochure | Simulator | Standard capacity |
|---|---|---|---|
| Primary | 575 st/hr | 576 st/hr | 481 st/hr |
| Secondary | 635 st/hr | 735 st/hr | 809 st/hr |
| Tertiary | 350 st/hr | 390 st/hr | 329 st/hr |

If the mesh size on the lower deck of the secondary screen is increased to 2", the overload on this screen is removed, and the simulation is close to the performance claimed by the vendor. Further refinements would be possible using the Karra screen model.

Note that the three sections of the plant are designed to handle different tonnages. Two stream splitters are included in the simulation to simulate this effect.

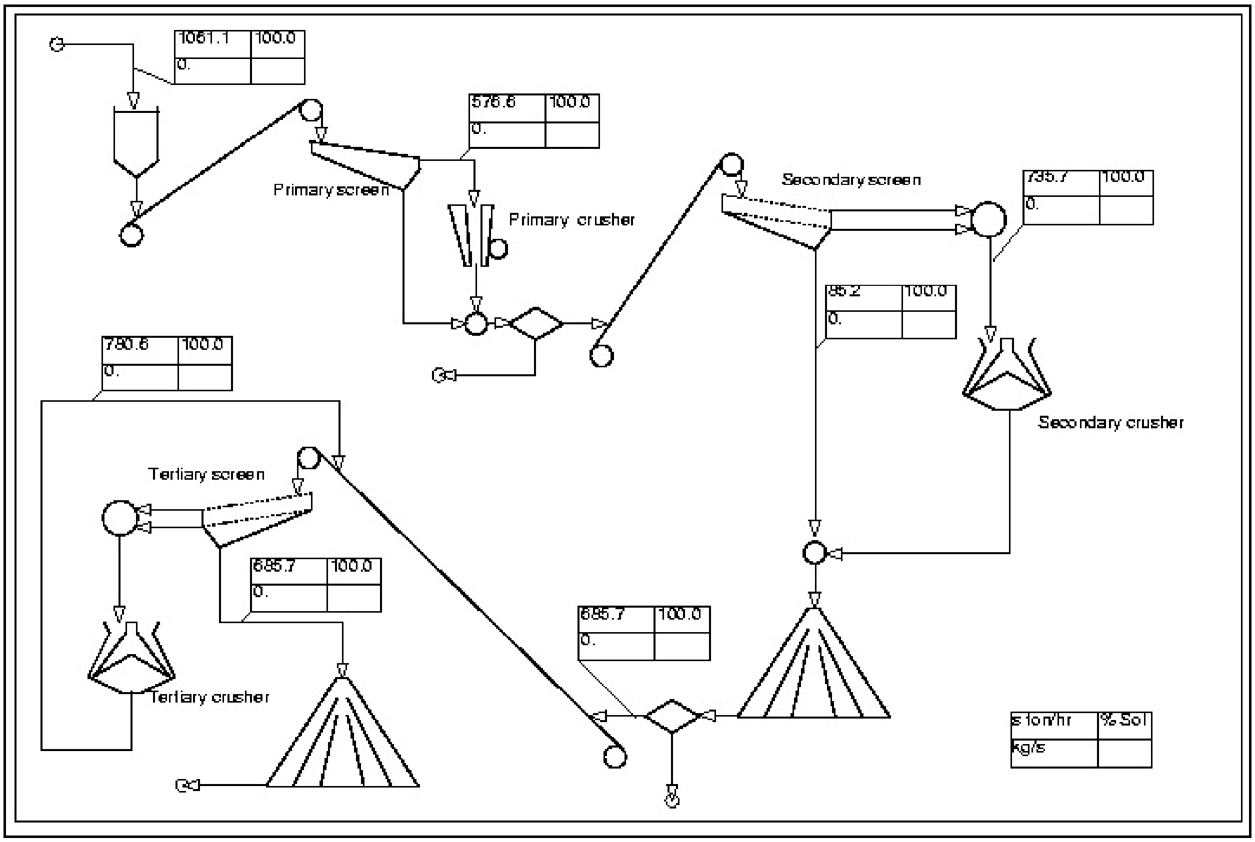


**Figure 1** Flowsheet A: A three stage crusher circuit.

## Some suggestions for the solution of Assignment 3-2

Look initially at the effect of variations in the closed-side setting of the secondary crusher.

Table 1 Variation of equipment performance and product rates as closed-side setting of the secondary crusher varies.

| CSS in secondary crusher mm | Circulating flow tonnes/hr | Secondary crusher power kW | Primary screen AUF Upper | Primary screen AUF Lower | Secondary screen AUF Fine | Secondary screen AUF Coarse | Fines production rate tonnes/hr | Coarse stone production rate tonnes/hr |
|---|---|---|---|---|---|---|---|---|
| 63 | 906.5 | 32.9 | 1.01 | 1.32 | 1.33 | 1.03 | 31.0 | 237.3 |
| 61.5 | 533.9 | 39.1 | 0.94 | 1.26 | 1.24 | 1.03 | 28.0 | 245.1 |
| 60 | 401.8 | 35.8 | 0.90 | 1.25 | 1.22 | 1.04 | 27.0 | 249.0 |
| 57 | 306.8 | 35.2 | 0.85 | 1.24 | 1.21 | 1.04 | 26.6 | 250.7 |
| 50 | 215.1 | 37.0 | 0.75 | 1.25 | 1.22 | 1.04 | 27.2 | 248.9 |
| 40 | 171.4 | 43.3 | 0.66 | 1.30 | 1.29 | 1.03 | 29.6 | 242.5 |
| 30 | 156.8 | 57.5 | 0.59 | 1.37 | 1.41 | 1.04 | 33.7 | 224.6 |
| 25 | 155.1 | 69.2 | 0.58 | 1.42 | 1.50 | 1.06 | 36.8 | 205.5 |
| 20 | 154.5 | 84.4 | 0.57 | 1.47 | 1.61 | 1.11 | 40.6 | 175.9 |
| 15 | 154.4 | 107.0 | 0.57 | 1.54 | 1.78 | 1.06 | 46.2 | 150.1 |

The dominant variation is found in the circulating flow in the closed crusher loop. The circulating flow is shown as a function of the closed-side setting in Figure 1. The installed crusher is too large for the duty required and a smaller crusher could have been specified in the original design. The rated capacity of different crusher is also shown and this makes it easy to

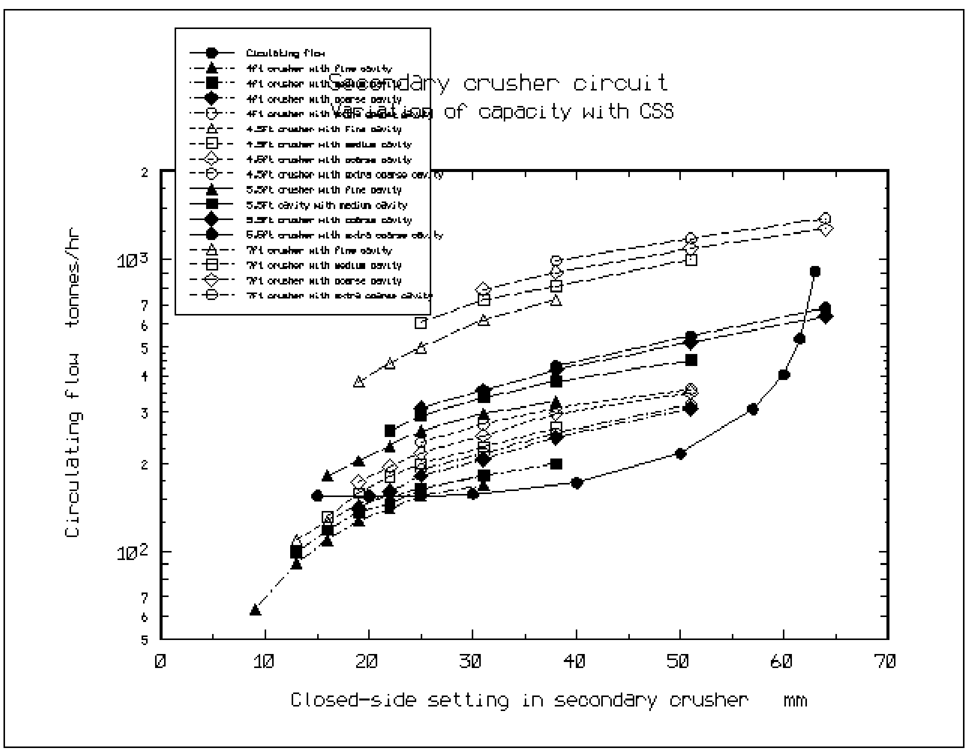


**Figure 2** Variation of the circulating load with close-side setting in the secondary crusher.

choose an appropriate crusher setup.
The production rate of fine and coarse products are shown in Figure 2. The production rates of these two products are not influenced by settings in the tertiary crusher circuit. The variation of power consumption in the secondary crusher is also shown as a function of the closed-side setting in Figure 2.

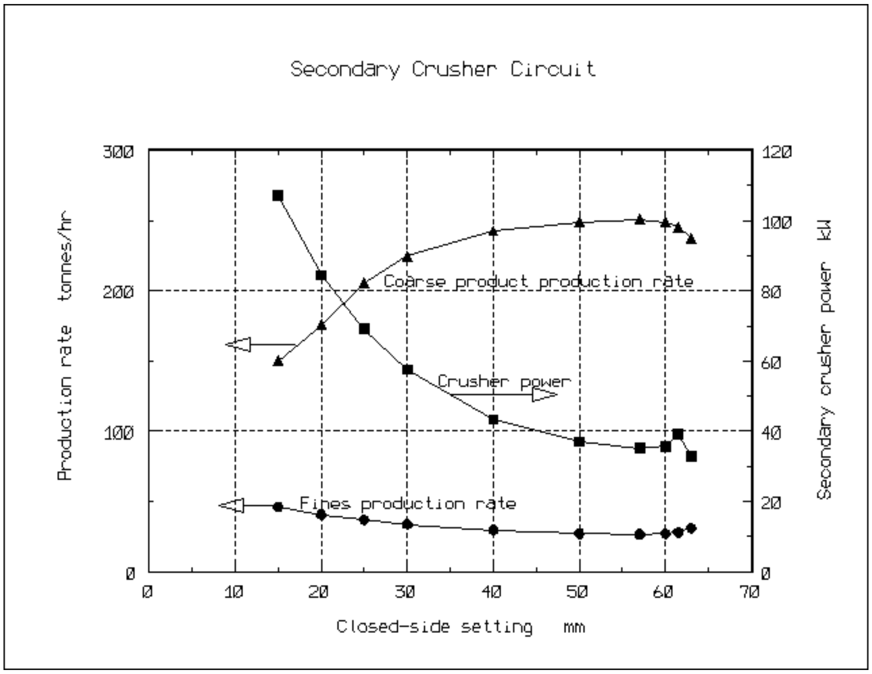


**Figure 3** Variation of production rates and crusher power demand with closed-side setting.

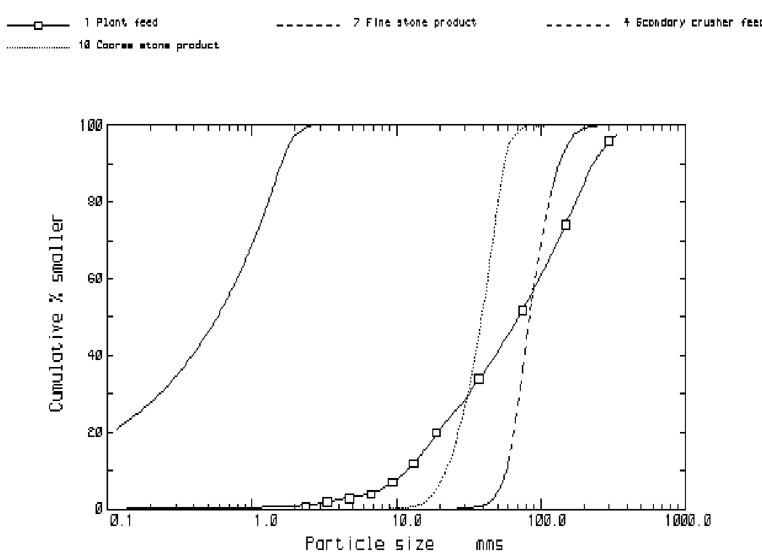


**Figure 4** Size distributions in the fine and coarse products with the secondary crusher set at 15mm.

The size distributions in the coarse and fine products are shown in Figures 3 and 4 at the extreme ends of the closed-side setting range.  These size distributions are determined rpimarily by the screens and so the variatiion of distribution of size is minimal as the closed-side setting changes.

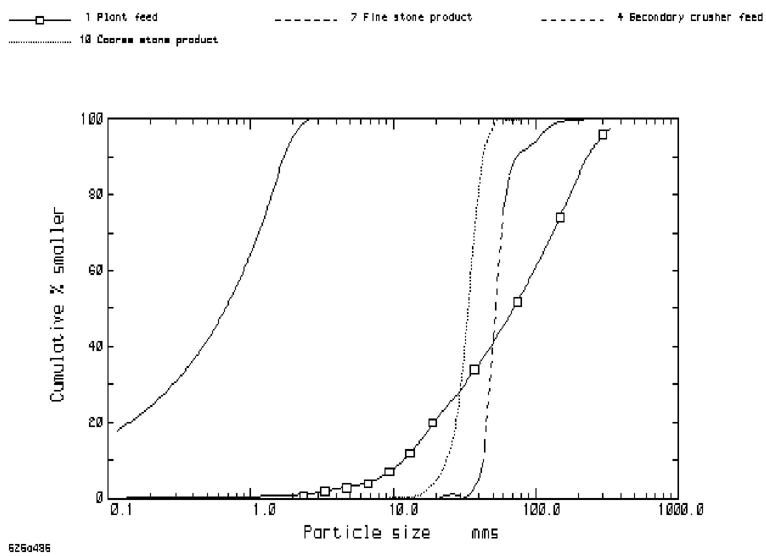


**Figure 5** Size distributions in coarse and fine products with the secondary crusher set at 63mm.